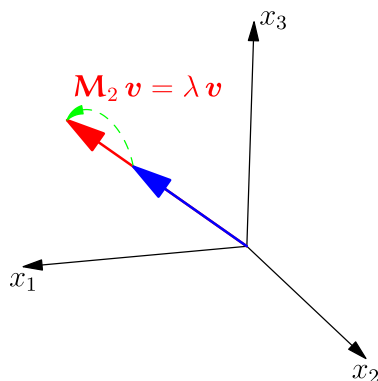
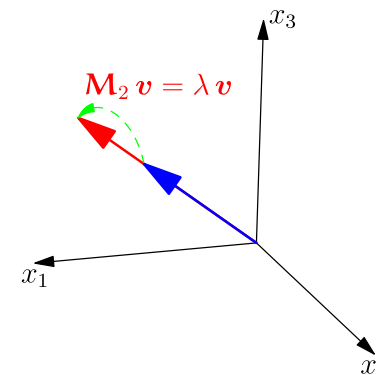


Eigensystems



Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank

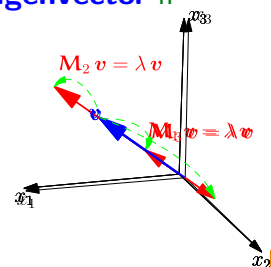
1. **Eigenvectors**
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation
5. Non-Normal Matrices



Eigenvector equation

- Eigen-systems help us to understand mappings
- A non-zero vector v is said to be an **eigenvector** if

$$Mv = \lambda v$$



- M is square (i.e. $n \times n$)
- Where the number λ is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

Symmetric Matrices

- If M is an $n \times n$ **symmetric** matrix then it has n real orthogonal eigenvectors with real eigenvalues
- We denote the i^{th} eigenvector by v_i and the corresponding eigenvalue by λ_i so that

$$Mv_i = \lambda_i v_i$$

- Orthogonal means that if $i \neq j$ then

$$v_i^T v_j = 0$$

- (We can always normalise eigenvectors if we want)

Proof of Orthogonality

- $(\mathbf{M}v_i = \lambda_i v_i)^\top$ implies $v_i^\top \mathbf{M}^\top = \lambda_i v_i^\top$
- When $\mathbf{M}$ is symmetric then $\mathbf{M}v_i = \lambda_i v_i \Rightarrow v_i^\top \mathbf{M} = \lambda_i v_i^\top$
- Consider two eigenvectors v_i and v_j of $\mathbf{M}$

$$\begin{aligned} v_i^\top \mathbf{M} v_j &= (v_i^\top \mathbf{M}) v_j = \lambda_i v_i^\top v_j \\ &= v_i^\top (\mathbf{M} v_j) = \lambda_j v_i^\top v_j \end{aligned}$$

- So either $\lambda_i = \lambda_j$ or $v_i^\top v_j = 0$
- If $\lambda_i = \lambda_j$ then any linear combination of v_i and v_j is an eigenvector $(\mathbf{M}(av_i + bv_j) = \lambda_i(av_i + bv_j))$. So I can choose two eigenvectors that are orthogonal to each other.

Orthogonal Matrices

- We can construct an **orthogonal** matrix $\mathbf{V}$ from the normalised eigenvectors ($v_i^\top v_i = 1$)

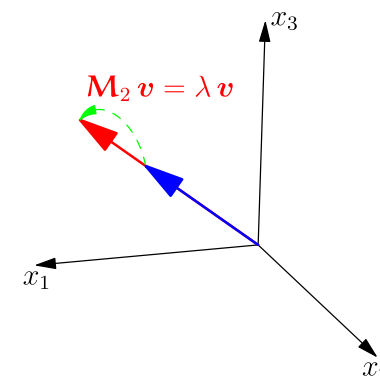
$$\mathbf{V} = (v_1, v_2, \dots, v_n)$$

- Matrix $\mathbf{V}$ is an $n \times n$ matrix
- Because the vectors v_i are orthogonal and normalised

$$\mathbf{V}^\top \mathbf{V} = \begin{pmatrix} v_1^\top v_1 & v_1^\top v_2 & \dots & v_1^\top v_n \\ v_2^\top v_1 & v_2^\top v_2 & \dots & v_2^\top v_n \\ \vdots & \vdots & \ddots & \vdots \\ v_n^\top v_1 & v_n^\top v_2 & \dots & v_n^\top v_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = \mathbf{I}$$

Outline

1. Eigenvectors
2. **Orthogonal Matrices**
3. Eigen Decomposition
4. Low Rank Approximation
5. Non-Normal Matrices



The Other Way Around

- We have shown that $\mathbf{V}^\top \mathbf{V} = \mathbf{I}$
- Thus multiply both sides on the left by $\mathbf{V}$

$$\mathbf{V} \mathbf{V}^\top \mathbf{V} = \mathbf{V} \mathbf{I}$$

- $\mathbf{V}$ will have an inverse (we show this on the next slide), $\mathbf{V}^{-1}$, such that $\mathbf{V} \mathbf{V}^{-1} = \mathbf{I}$
- Multiplying the equation on the right by $\mathbf{V}^{-1}$

$$\begin{aligned} (\mathbf{V} \mathbf{V}^\top) \mathbf{V} \mathbf{V}^{-1} &= \mathbf{V} \mathbf{V}^{-1} \mathbf{I} \\ \mathbf{V} \mathbf{V}^\top &= \mathbf{I} \end{aligned}$$

- Note that, $\mathbf{V}^{-1} = \mathbf{V}^\top$ (definition of orthogonal matrix)

Invertible Matrices

- A matrix, $\mathbf{M}$, will be singular (not invertible) if there exists a vector $\mathbf{x} (\neq \mathbf{0})$ such that

$$\mathbf{M}\mathbf{x} = \mathbf{0}$$

- Now if there exists such a vector such that $\mathbf{V}\mathbf{x} = \mathbf{0}$ then multiplying by $\mathbf{V}^T$ we get

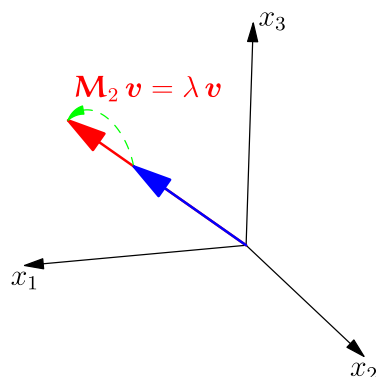
$$\begin{aligned}\mathbf{V}^T\mathbf{V}\mathbf{x} &= \mathbf{V}^T\mathbf{0} \\ \mathbf{x} &= \mathbf{0}\end{aligned}$$

since $\mathbf{V}^T\mathbf{V} = \mathbf{I}$

- Thus $\mathbf{V}$ is invertible

Outline

- Eigenvectors
- Orthogonal Matrices
- Eigen Decomposition**
- Low Rank Approximation
- Non-Normal Matrices



Rotations

- Orthogonal matrices satisfy $\mathbf{V}^T\mathbf{V} = \mathbf{V}\mathbf{V}^T = \mathbf{I}$
- As a consequence they define rotations (and possibly a reflection)
- Consider a vector $\mathbf{x}$ and $\mathbf{x}' = \mathbf{V}\mathbf{x}$, now

$$\|\mathbf{x}'\|_2^2 = \mathbf{x}'^T\mathbf{x}' = (\mathbf{V}\mathbf{x})^T(\mathbf{V}\mathbf{x}) = \mathbf{x}^T\mathbf{V}^T\mathbf{V}\mathbf{x} = \mathbf{x}^T\mathbf{x} = \|\mathbf{x}\|_2^2$$

- Similarly if additionally $\mathbf{y}' = \mathbf{V}\mathbf{y}$ then

$$\langle \mathbf{x}', \mathbf{y}' \rangle = (\mathbf{V}\mathbf{x})^T(\mathbf{V}\mathbf{y}) = \mathbf{x}^T\mathbf{V}^T\mathbf{V}\mathbf{y} = \mathbf{x}^T\mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|_2\|\mathbf{y}\|_2\cos(\theta)$$

- Rotations and reflections preserve lengths and angles

Matrix Decomposition

- Taking the matrix of eigenvectors, $\mathbf{V}$, then

$$\mathbf{M}\mathbf{V} = \mathbf{M}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n) = (\lambda_1\mathbf{v}_1, \lambda_2\mathbf{v}_2, \dots, \lambda_n\mathbf{v}_n) = \mathbf{V}\mathbf{\Lambda}$$

- where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$

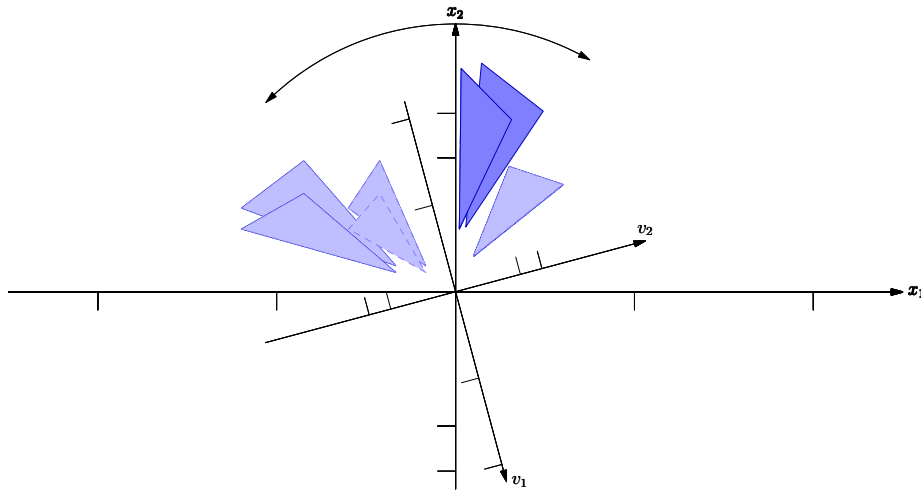
- Now

$$\mathbf{M} = \mathbf{M}\mathbf{V}\mathbf{V}^T = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

- Very important *similarity transform*

Mappings by Symmetric Matrices

$$\mathbf{M} = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \begin{pmatrix} \cos(-75) & \sin(-75) \\ -\sin(-75) & \cos(-75) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75) & \sin(75) \\ -\sin(75) & \cos(75) \end{pmatrix}$$



Inverses

- For any symmetric invertible matrix

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T \quad \mathbf{M}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$$

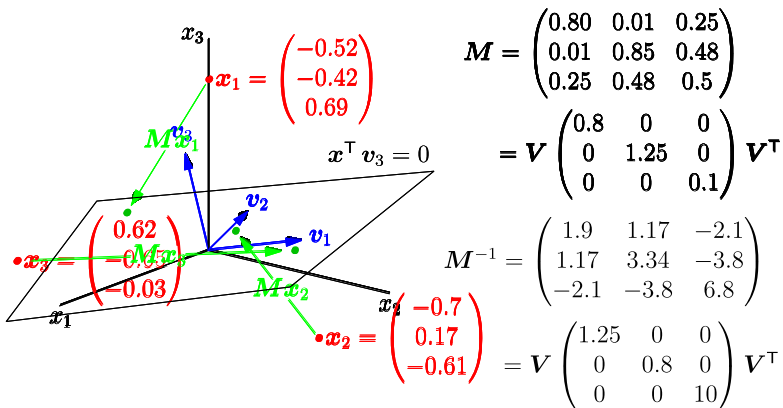
- Where $\mathbf{\Lambda}^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$\begin{aligned} \mathbf{M}\mathbf{M}^{-1} &= (\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T)(\mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T) = \mathbf{V}\mathbf{\Lambda}(\mathbf{V}^T\mathbf{V})\mathbf{\Lambda}^{-1}\mathbf{V}^T \\ &= \mathbf{V}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{V}^T = \mathbf{V}\mathbf{V}^T = \mathbf{I} \end{aligned}$$

- i.e. small eigenvalues become large eigenvalues and vice versa

Ill-Conditioning Again



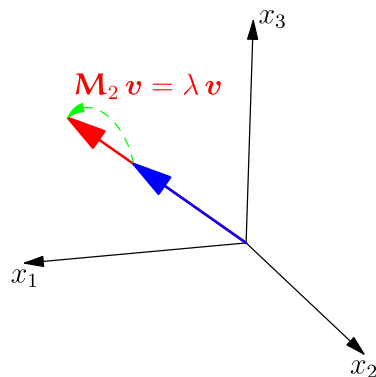
Condition Number

- Taking matrix inverses can be inherently unstable
- Any small error can be amplified by taking the inverse
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)
- For symmetric matrices the largest $|\lambda|$ is the spectral norm, $\|\mathbf{M}\|_2$, from the vector spaces lecture
- The condition number is given by

$$\|\mathbf{M}\|_2 \times \|\mathbf{M}^{-1}\|_2 = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

- Large condition number implies very ill-conditioned

1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. **Low Rank Approximation**
5. Non-Normal Matrices



“Inverting” Rank Deficient Matrices

- Rank deficient matrices are non-invertible (i.e. we don't know the vector x such that $Mx = b$) as we don't know the component of the x in the null space
- Although we don't know x we can find a vector, x , that satisfies $Mx = b$ (provided one exists)
- Given a symmetric $n \times n$ matrix with k non-zero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ we can construct a “pseudo-inverse” M^+ as $V\Lambda^+V^T$ where $\Lambda^+ = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0)$
- This finds the vector x with no component in the null space (it is the solution with the smallest norm)
- This and $(X^TX)^{-1}X^T$ are special cases of the pseudo-inverse we meet with SVD

- The rank of a symmetric matrix, M , is the number of non-zero eigenvalues
- The space spanned by the eigenvectors v_a, v_b , etc. with zero eigenvalue forms a **null space**
- Any vector in the null space will get mapped to the zero vector

$$M(av_a + bv_b + \dots) = 0$$

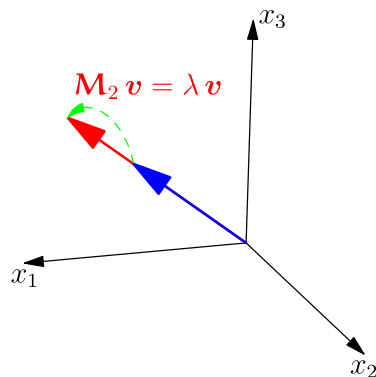
- A square matrix is said to be **rank deficient** if it has any eigenvectors with eigenvalue equal to 0
- This happens when the columns of the matrix are not linearly independent

Low Rank Approximation

- Recall that matrices with large and small eigenvalues are ill-conditioned so the inverse has the potential to greatly amplify any measurement error
- One work around is to set all small eigenvalues to zero and use the pseudo-inverse
- Setting small eigenvalues to zero reduces the rank of the matrix and is an example of a low rank approximation
- Low rank approximations are much used to obtain approximate models for arrays of data (we will revisit this when we look at SVD)

Outline

1. Eigenvectors
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation
5. **Non-Normal Matrices**



Non-Normal Matrices

- Symmetric matrices are part of a class of matrices known as **normal matrices**, which satisfy $\mathbf{M}\mathbf{M}^\dagger = \mathbf{M}^\dagger\mathbf{M}$ (for real matrices $\mathbf{M}^\dagger = \mathbf{M}^T$)
- Hermitian matrices, $\mathbf{H}$, are complex matrices that equal their conjugate transpose: $\mathbf{H} = (\mathbf{H}^T)^* = \mathbf{H}^\dagger$
- Hermitian matrices are also normal; they are important in quantum theory, less so in machine learning
- Normal matrices are exactly those with a full set of orthonormal (possibly complex) eigenvectors
- Most non-symmetric real matrices are not normal (rotations are an exception); their eigenvectors are more confusing

Asymmetric Matrices

- For $n \times n$ real symmetric matrices we are guaranteed there are n orthogonal real eigenvectors
- For non-symmetric matrices the eigenvectors will generally not be orthogonal, they won't necessarily be real
- We are not even guaranteed there will be n linearly independent eigenvectors
- Matrices which don't have n linearly independent eigenvectors are said to be **defective**, e.g. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one
- Randomly generated matrices will typically not be defective, but nevertheless they can arise in certain situations

Characteristic Equation

- Since our eigenvectors are defined by $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ then $(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ has a solution $\mathbf{v} \neq \mathbf{0}$ only if $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$
- This is known as the characteristic equation and leads to an n^{th} order polynomial in λ
- An n^{th} order polynomial has n roots (counting repeats), which are in general complex
- Thus even for real (non-symmetric) matrices we can get complex eigenvalues

Complex Conjugate Pairs

- If $\mathbf{M}$ is real and λ is complex then $\mathbf{v}$ must be complex too, otherwise $\mathbf{M}\mathbf{v}$ would be real but $\lambda\mathbf{v}$ would not

- Furthermore

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v} \quad \implies \quad \mathbf{M}\mathbf{v}^* = \lambda^*\mathbf{v}^*$$

- So that eigenvectors and their corresponding eigenvalues exist in complex conjugate pairs

Left and Right Eigenvectors

- For symmetric matrices if $\mathbf{v}$ is a right-hand eigenvector in the sense that $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v}^T$ is a left-hand eigenvector

$$\mathbf{v}^T\mathbf{M} = \lambda\mathbf{v}^T$$

- Earlier we used this property to show that the eigenvectors for symmetric matrices are orthogonal
- For asymmetric real matrices the left-hand eigenvectors $\mathbf{u}^T$ are different from the right-hand eigenvectors, although they share the same eigenvalues (they have the same characteristic equation)

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v} \quad \mathbf{u}^T\mathbf{M} = \lambda\mathbf{u}^T$$

- As before, $\mathbf{u}_i^T\mathbf{v}_j = 0$ if $\lambda_i \neq \lambda_j$, but in general $\mathbf{v}_i^T\mathbf{v}_j \neq 0$

Eigen-Decomposition

- For a non-symmetric matrix, $\mathbf{M}$, provided it is not defective, we can decompose the matrix as

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$$

- $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues
- $\mathbf{V}$ is a matrix whose columns are the right-hand eigenvectors—in general it is not an orthogonal matrix
- $\mathbf{V}^{-1}$ is a matrix whose rows are the left-hand eigenvectors (scaled so that $\mathbf{u}_i^T\mathbf{v}_i = 1$)

Summary

- Linear mappings are commonly used in machine learning algorithms such as regression
- We can understand symmetric operators by looking at their eigenvectors
- Any symmetric matrix can be decomposed as $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
 - ★ where $\mathbf{V}$ is an orthogonal matrix whose columns are the eigenvectors
 - ★ and $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues
- This decomposition allows us to understand inverse mappings
- Non-symmetric matrices are more complex—we use the eigen-decomposition less because of this